discussed in Chapter 12—might destroy a wormhole just as it was getting into position to be a time machine. The reason is that, just at the moment when time travel through the wormhole becomes possible, a devastating feedback mechanism, somewhat like the screeching noise generated when microphone and speaker levels in a sound system are not adjusted appropriately, may come into play. Vacuum fluctuations from the future can travel through the wormhole to the past, where they can then travel through ordinary space and time to the future, enter the wormhole, and travel back to the past again, creating an endless cycle through the wormhole and filling it with ever-increasing energy. Presumably, such an intense energy buildup would destroy the wormhole. Theoretical research suggests this as a real possibility, but the necessary calculations strain our current understanding of general relativity and quantum mechanics in curved spacetime, so there is no conclusive proof.

The challenges to building a wormhole time machine are clearly immense. But the final word won't be given until our facility with quantum mechanics and gravity is refined further, perhaps through advances in superstring theory. Although at an intuitive level physicists generally agree that time travel to the past is impossible, as of today the question has yet to be fully closed.

Cosmic Rubbernecking

In thinking about time travel, Hawking has raised an interesting point. Why, he asks, if time travel is possible, haven't we been inundated with visitors from the future? Well, you might answer, maybe we have. And you might go further and say we've put so many time travelers in locked wards that most of the others don't dare identify themselves. Of course, Hawking is half joking, and so am I, but he does raise a serious question. If you believe, as I do, that we have not been visited from the future, is that tantamount to believing time travel impossible? Surely, if people succeed in building time machines in the future, some historian is bound to get a grant to study, up close and personal, the building of the first atomic bomb, or the first voyage to the moon, or the first foray into reality television. So, if we believe no one has visited us from the future,